



“The Fog of Crisis” – Strategic Complexity in Executive Communication

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- 1 Research Question & Hypothesis**
- 2 Descriptive Evidence: The COVID Crisis Shock**
- 3 Model 1: Overall Effect of COVID on Communication Complexity**
- 4 Model 2: Industry Vulnerability and Differential Effects**
- 5 Conclusion**



Did COVID-19 (2020) increase executive communication complexity?

- **Research Question:**

Did the onset of the COVID-19 pandemic cause executives to use significantly more complex language (higher Gunning Fog Index) compared to pre-pandemic trends? Was this effect uniform, or stronger in COVID-vulnerable industries?

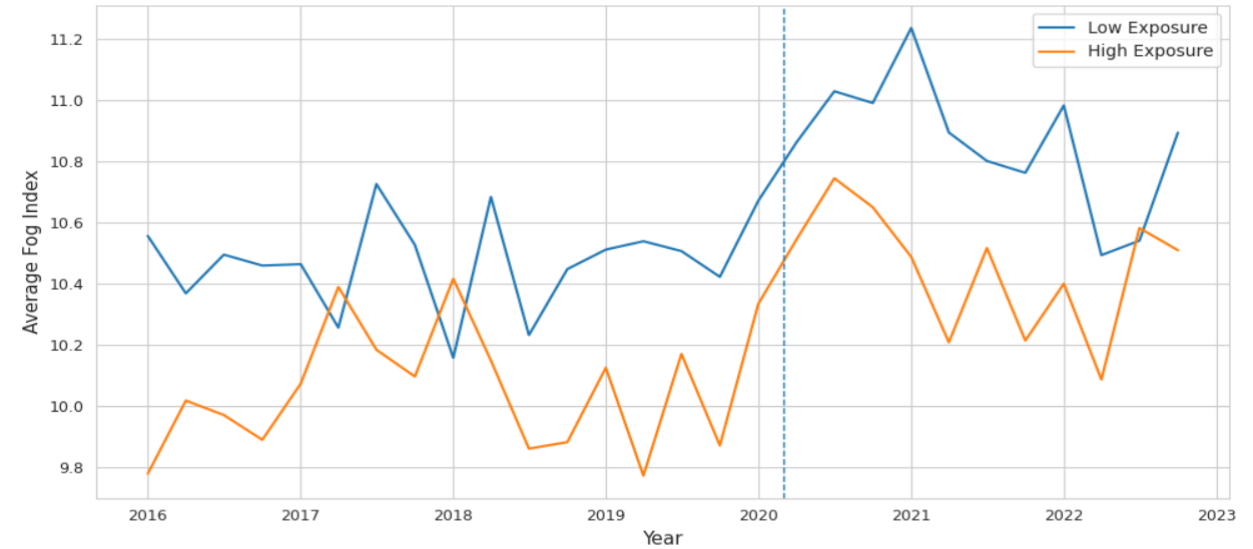
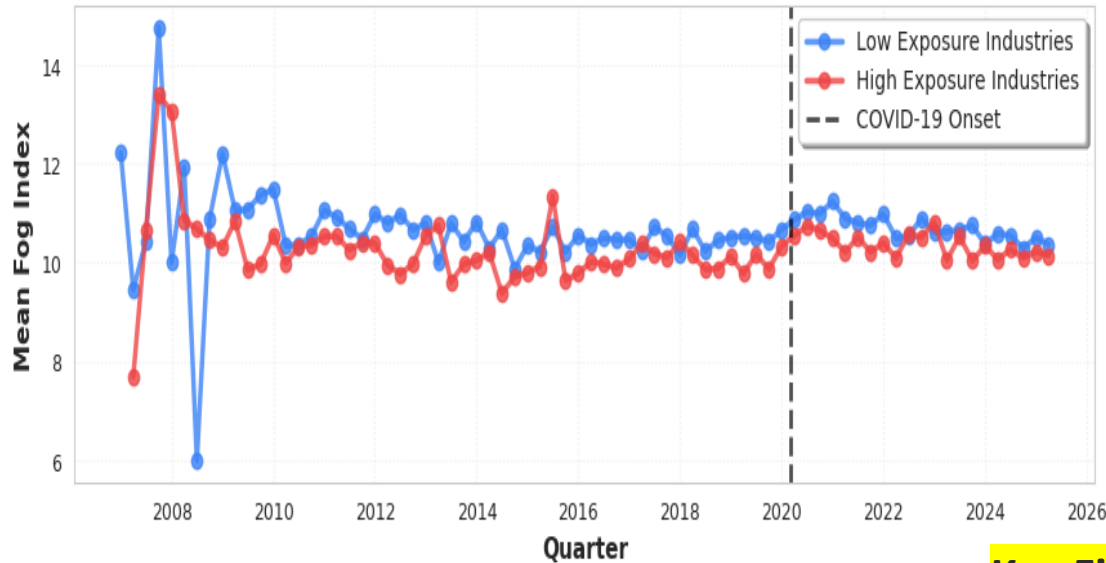
- **Hypothesis:**

During the 2020 crisis, executive communication complexity increased significantly relative to historical baselines, reflecting defensive vagueness, which is a strategic use of longer sentences and complex vocabulary to navigate extreme uncertainty and limit market panic.

Descriptive Evidence: The COVID Crisis Shock



Fig 1: Parallel Trends Analysis - Fog Index Over Time



Key Findings

- **Similar pre-COVID trends:** High- and low-exposure industries follow nearly identical Fog Index paths before 2020, supporting comparability.
- **Clear COVID-era increase:** Both groups show a visible upward shift in communication complexity after the pandemic onset.
- **No strong divergence:** High-exposure industries do not display a systematically larger increase than resilient industries.
- **Interpretation:** The rise in complexity reflects a **broad crisis-driven response**, rather than sector-specific obfuscation.

High Exposure	COVID Period	Mean Fog Index	Count
0	0	10.594	2939
	1	10.750	2647
1	0	10.181	1981
	1	10.424	2147

Model 1: Overall Effect of COVID on Communication Complexity



Model

$$FogIndex_{i,t} = \beta_1 COVID_t + \beta' X_{i,t} + \alpha_i + \varepsilon_{i,t}$$

Key components:

$FogIndex_{i,t}$: Disclosure complexity of firm i at time t

$COVID_t$: Indicator equal to 1 for the post-2020 period

$X_{i,t}$: Vector of control variables (sentiment, language features, and video characteristics)

α_i : Firm fixed effects

$\varepsilon_{i,t}$: Error term, with standard errors clustered at the firm level

Parameter Estimates

	Parameter	Std. Err.	T-stat	P-value	Lower CI	Upper CI
covid	0.4593	0.0655	7.0150	0.0000	0.3310	0.5877
neg_ratio	7.3060	5.4113	1.3502	0.1770	-3.3013	17.913
pos_ratio	33.710	3.8873	8.6718	0.0000	26.090	41.330
uncert_ratio	6.0938	6.1200	0.9957	0.3194	-5.9030	18.090
exec_age	-0.0136	0.0051	-2.6417	0.0083	-0.0237	-0.0035
exec_tenure	-0.0091	0.0088	-1.0299	0.3031	-0.0263	0.0082
is_CEO	-0.2148	0.0953	-2.2540	0.0242	-0.4016	-0.0280
log_duration	-0.1203	0.0301	-3.9910	0.0001	-0.1794	-0.0612
log_views	-0.1069	0.0181	-5.9135	0.0000	-0.1423	-0.0714
stock_return	0.0837	0.1147	0.7302	0.4653	-0.1411	0.3085
volatility	0.0282	0.0860	0.3274	0.7433	-0.1405	0.1968

Key Findings

- **COVID-19 significantly increased disclosure complexity.**
- Fog Index rose by about **0.45 points** after **COVID onset** ($p < 0.001$).
- Indicates firms used **more complex and less readable language** during the pandemic.
- Effect is **robust after controlling for firm traits, sentiment, attention, and risk factors.**

Model 2: Industry Vulnerability and Differential Effects



Model

$$\text{FogIndex}_{i,t} = \beta_1(\text{COVID}_t \times \text{HighExposure}_i) + \beta'X_{i,t} + \alpha_i + \delta_t + \varepsilon_{i,t}$$

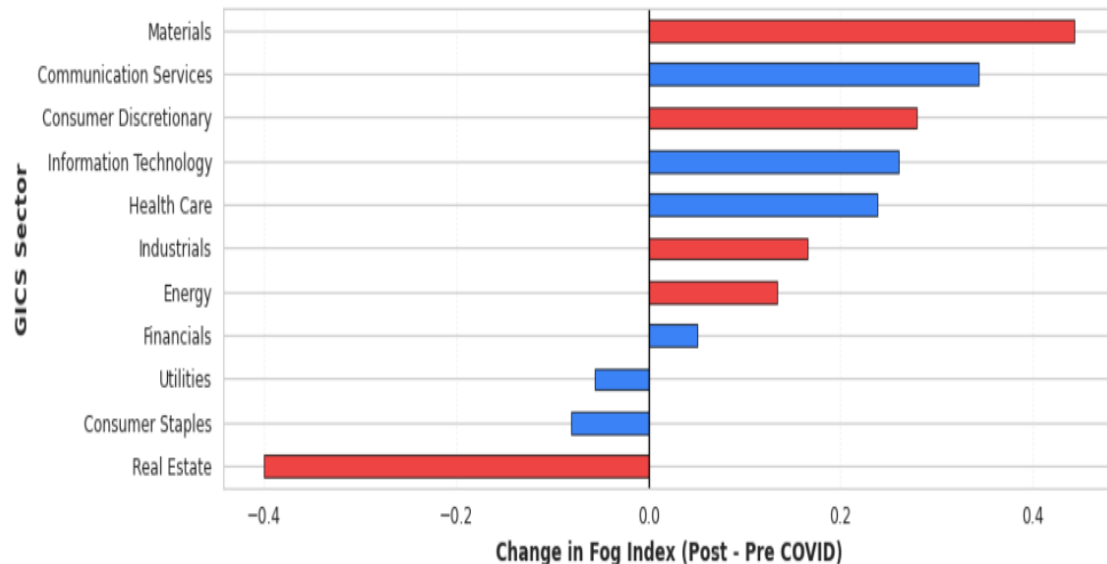
Where:

HighExposure_i : Indicator equal to 1 if firm *i* belongs to a high-exposure industry

COVID_t × HighExposure_i : Difference-in-differences treatment term

δ_t : Time (quarter) fixed effects

Fig 2: Fog Index Change by Sector (Red = High Exposure)



Parameter Estimates

	Parameter	Std. Err.	T-stat	P-value	Lower CI	Upper CI
covid_x_exposure	0.0196	0.1109	0.1764	0.8600	-0.1978	0.2369
neg_ratio	7.4519	5.6846	1.3109	0.1899	-3.6914	18.595
pos_ratio	34.132	4.0739	8.3782	0.0000	26.146	42.118
uncert_ratio	5.6937	6.5388	0.8708	0.3839	-7.1239	18.511
exec_age	-0.0084	0.0059	-1.4266	0.1537	-0.0200	0.0031
exec_tenure	0.0040	0.0105	0.3835	0.7014	-0.0165	0.0246
is_CEO	-0.1641	0.1185	-1.3844	0.1663	-0.3964	0.0682
CEO_Duality	-0.1329	0.1189	-1.1177	0.2637	-0.3661	0.1002
log_duration	-0.1081	0.0323	-3.3435	0.0008	-0.1714	-0.0447
log_views	-0.1091	0.0197	-5.5368	0.0000	-0.1477	-0.0704
stock_return	0.1102	0.1227	0.8985	0.3690	-0.1303	0.3507
volatility	0.0014	0.0921	0.0149	0.9881	-0.1791	0.1819
roa	0.3216	0.2375	1.3538	0.1758	-0.1441	0.7872

Key Findings

- **No significant difference** between high- and low-exposure industries after COVID
- Disclosure complexity increase is **broad-based across sectors**
- Sector changes vary, but **not statistically strong**
- **Call features (duration, views) matter more than industry exposure**



- Executive communication became **more complex after the onset of COVID-19.**
- This increase occurred **across most firms and industries**, indicating a common response to crisis-related uncertainty.
- Industries that were more severely affected by COVID **did not** show a consistently larger increase in complexity once broader crisis effects are taken into account.
- Overall, the findings suggest that increased complexity reflects a **general reaction to heightened uncertainty**, rather than deliberate obfuscation by specific industries.



- [1] Gunning, R. (1969). The fog index after twenty years. *Journal of Business Communication*, 6(2), 3-13.

- [2] Bloomfield, R. J. (2002). The 'incomplete revelation hypothesis' and financial reporting.

- [3] Loughran, T., & McDonald, B. (2014). Measuring readability in financial disclosures. *the Journal of Finance*, 69(4), 1643-1671.